

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6EJ303 (2)			
Course Title	APPLIED PROBABILITY			
Type of Course	Elective (Specialisation- Data Science)			
Semester	VI			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic Algebra and Calculus			
Course Summary	This course serves as an introduction to the fundamental principles and concepts of probability theory. Understanding probability distributions, expectations, and Markov chains is essential for modelling data, making predictions, and analysing complex systems in data science applications.			

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand basic concepts in probability theory, including discrete and continuous probability distributions, joint distributions for multiple random variables, and Markov chains.	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Apply probability distributions to practical scenarios and compute key measures such as expected value and variance, with an emphasis on their significance in decision-making and risk assessment.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Explore and understand fundamental limit theorems, such as the law of large numbers and the central limit theorem, and their implications for probability theory and statistical inference.	U	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		Introduction to Probability Models - Sheldon M Ross -10 th (e)- Academic Press		
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I			12	
	1	Sample space and events.		Min 15
	2	Probabilities defined on events.		
	3	Conditional Probabilities		
	4	Independent Events.		
	5	Bayes 'Formula.		
	6	Random Variables.		
	7	Discrete Random Variables.		
	8	Continuous Random Variables		
	Chapter 1: Sections 1.2, 1.3, 1.4, 1.5, 1.6 Chapter 2: Sections 2.1, 2.2, 2.3			
II			12	
	9	Expectation of a Random Variable – Discrete Case and Continuous Case		Min 15
	10	Jointly distributed Random Variables.		
	11	Moment generating functions.		
	12	Limit Theorems		
	Chapter 2: sections 2.4, 2.5, 2.6, 2.8			
III			12	
	13	Conditional probability and conditional expectation- The discrete case.		Min 15
	14	Conditional probability and conditional expectation- The continuous case.		
	15	Computing expectations by conditioning.		
	16	Computing Probabilities by conditioning.		
	Chapter3: Sections 3.1, 3.2, 3.3, 3.4, 3.5			
IV			12	
	19	Markov chain – definition and examples.		

	20	Chapman-Kolmogrov equations.		Min 15
	21	Classification of states of a Markov Chain.		
	22	Limiting Probabilities		
	Chapter4: Sections 4.1, 4.2, 4.3, 4.4			
V	Open Ended		12	
	23	Properties of exponential distribution, Counting processes, Poisson process, properties of Poisson process		
References:				
1. S. Ross, “A First Course in Probability,” Eighth Edition, Prentice Hall. 2. W. Feller, “An Introduction to Probability Theory and its Applications,” Vol.I, John Wiley. 3. B.V. Gnedenko, “Theory of Probability,” Chelsea, New York 4. S.M. Ross, “Stochastic Processes,” second edition, John Wiley 5. S. Karlyn and H. Taylor, “A First course in Stochastic Processes”, second edition, Academic Press				

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	1	2	3	2	3	2	3	1	2
CO 2	2	3	2	2	3	2	3	3	3	1	3
CO 3	3	2	1	2	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓